GenCC Recommendation for Loss of Function (LOF) mechanism curation - Version 1.0

The Gene Curation Coalition proposes a semi-quantitative metric to determine the mechanism of disease for gene-disease relationships. The framework is intended to be used for human gene-disease relationships that have scored at Moderate or higher using the ClinGen gene curation framework (Strande et al 2017, see here for the most recent gene curation SOP). Levels should be capped at "Suspected" if applying to genes with Limited or below evidence. Caution should be exercised if adapting to non-human genetic diseases as this framework has not been tested using non-human phenotypes.

This framework is for evaluating genes for a monoallelic or biallelic loss of function mechanism.

**While this is a semi-quantitative framework, please note that points are not statistical in nature, nor do they correspond to Bayesian points in the sequence variant classification frameworks.

Table 1. Scoring Matrix for Biallelic Loss of Function Framework

Variant Type[1]	Evidence	Points
pLOF spectrum	At least 90 unique pLOF (>30% of P/LP variants are pLOF) and P/LP pLOF variants found in at least 3 distinct exons OR LOF represents at least ⅓ of all LP/P variants with at least 8 variants observed	0.9 pts[2] OR 0.3pts
pLOF Variant	Meets PVS1 (ignoring mechanism requirement; variants should be NMD+)	0.2 pts/variant (+0.1 if additional functional evidence of type below)
non-pLOF variants meeting P/LP[3]	Functional variant-level evidence demonstrating absence/degradation of transcript or protein	0.15 pt (range 0.1 - 0.2)
Experimental Evidence	KO model organism with consistent phenotype and zygosity	0.3 pt maximum cap (range 0.1-0.3)
Consistency in population databases	Absence or significantly reduced presence of homozygous variants of the type above,	0.15 pts

	considering the disease severity, age of onset and known presence of affected individuals in gnomAD from biobanks, etc	

Footnotes:
[1]Variants can only be counted when part of a case that has biallelic variants of the types above that are assumed to be causal based on consistent phenotype (cut points in half for a pLOF if the second variant is rare and predicted damaging but does not meet P/LP). Each variant can only be scored once; even if that variant has been observed in recurrent cases.
[2]Points should be given for only one pLOF spectrum scenario, not both. pLOF spectrum points should only be given to genes with one gene-disease relationship or if multiple, the variant type separation must be very clear.
[3]It is suggested that all pLOF variants be scored first and if this doesn't max out the framework, consider performing rapid variant curations and scoring non pLOF variants that have strong functional evidence and are likely pathogenic or pathogenic

Abbreviations: LOF, Loss of Function; PVS1 ACMG code see Abou Tayoun et al; NMD+, nonsense mediated decay is predicted to occur; pLOF, predicted LOF; P/LP, pathogenic/likely pathogenic; KO, knock out

Table 2. Scoring Matrix for Monoallelic Loss of Function Framework

Variant Type[1]	Evidence	Points
pLOF spectrum	At least 90 unique pLOF (>30% of P/LP variants are pLOF) and P/LP pLOF variants found in at least 3 distinct exons	0.9 pts[2]
pLOF variant	Meets PVS1 (ignoring mechanism requirement; variants should be NMD+)	0.2 pts/variant (+0.1 for functional evidence of type below)
non-pLOF variants meeting P/LP[3]	Functional variant-level evidence demonstrating absence/degradation of transcript or protein	0.15 pt (range 0.1 - 0.2)
Experimental evidence	KO model organism with consistent phenotype	Het KO 0.3 points Hom KO 0.15 points[3]
Consistency in population databases	Upper limit of o/e in gnomAD is <0.3 for early onset and <0.6 for late onset diseases	0.15 pts
	>/= 50% of pLOF variants with >1 alleles in gnomAD have a criteria-provided P/LP classification in ClinVar (min 3 variants needed)	0.1 pts

Footnotes:
[1]Variants can only be counted when the variant is assumed to be causal based on consistent phenotype and the disease inheritance is consistent with dominant or *de novo* occurrence. Each variant can only be scored once; even if that variant has been observed in recurrent cases.
[2]pLOF spectrum points should only be given to genes with one gene-disease relationship or if multiple, the variant type separation must be very clear.
[3]It is suggested that all pLOF variants be scored first and if this doesn't max out the framework, consider performing rapid variant curations and scoring non pLOF variants that have strong functional evidence and are likely pathogenic or pathogenic
[3]As it is not uncommon to only observe a human dominant phenotype in a homozygous state in an animal model, reduced points may be given for a homozygous knock out animal model. Only one zygosity (preferably affected heterozygous animals if available) should be scored
Abbreviations: LOF, Loss of Function; PVS1 ACMG code see Abou Tayoun et al; NMD+, nonsense mediated decay is predicted to occur; P/LP, pathogenic/likely pathogenic; pLOF, predicted LOF; KO, knock out; het, heterozygous; hom, homozygous; o/e, observed/expected

value in gnomAD

Table 3. Summary Matrix for Loss of Function Framework

Points from either Table 1 or Table 2 should be summed and the equivalent confidence term applied to the LOF mechanism.

ClinGen Dosage Sensitivity Score	Evidence Points	Confidence Term	Mechanism
3*	.99 or higher	Established	Loss of function (either monallelic or biallelic)
Any score lower than 3, evaluate mechanism with LOF frameworks	0.90-0.98	Likely	
	0.5-0.89	Suspected	
	0.49-0	Uncertain	

*A ClinGen dosage sensitivity score of 3 can be used as an equivalent for established monoallelic loss of function mechanism. For scores below 3, the mechanism should be evaluated using this framework